# Measure Theoretic Probability
## Lesson 25
### Expectation of Non-Negative Random Variables

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# Outline

1. Where We Are and the Plan
2. Definition for Non-Negative Random Variables
3. Approximating by Simple Random Variables
4. Expectation Passes Through the Limit
5. The Increasing Sets Claim
6. A Worked Example
7. Summary and Exercises

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# Recap: Expectation of a Simple Random Variable

## Setting

$(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. A **simple** random variable is

$$Y = \sum_{i=1}^{m} y_i \, I_{B_i}, \qquad B_i \in \mathcal{F},$$

(canonical form: $y_i$ distinct, $B_i = \{Y = y_i\}$ partition $\Omega$), and

$$\mathbb{E}[Y] = \sum_{i=1}^{m} y_i \, \mathbb{P}(B_i).$$

## Facts we will use (Lessons 23–24)

- $\mathbb{E}[Y]$ does not depend on the representation.
- Linearity, and **monotonicity**: $Y_1 \le Y_2 \Rightarrow \mathbb{E}[Y_1] \le \mathbb{E}[Y_2]$.
- $\mathbb{E}[I_B] = \mathbb{P}(B)$.

# The Goal Today

## Problem

Simple random variables take only finitely many values. A continuous random variable (say $X(\omega) = -2\ln\omega$) does not. How do we define $\mathbb{E}[X]$ **without** density functions?

## The idea, in one sentence

Approximate $X$ **from below** by simple random variables, whose expectations we already know, and take a limit (a supremum).

## Roadmap

1. Define $\mathbb{E}[X]$ for $X \geq 0$ as a supremum over simple $Y \leq X$.
2. Show that every $X \geq 0$ is an increasing limit of simple random variables $X_n \uparrow X$.
3. Show that $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$, so we can compute $\mathbb{E}[X]$ as a limit.
4. Worked example: $\mathbb{E}[-2\ln\omega] = 2$.

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# Definition

## Definition

Let $X : \Omega \to [0, \infty)$ be a random variable. Define

$$\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : \ Y \text{ simple random variable, } 0 \leq Y \leq X\} \ \in [0, \infty].$$

## Reading the definition

Look at **every** step function $Y$ that fits underneath $X$ (pointwise $0 \leq Y(\omega) \leq X(\omega)$). Compute each $\mathbb{E}[Y]$. The expectation of $X$ is the **largest** of these numbers (the supremum).

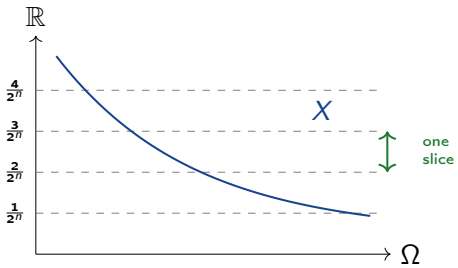
## Remarks

- The value may be $+\infty$ (the supremum of an unbounded set).
- If $X$ is itself simple, the supremum is attained at $Y = X$ (monotonicity), so the new definition agrees with the old one.
- If $X \equiv 0$, every admissible $Y$ is 0, so $\mathbb{E}[X] = 0$.
- The definition looks hard to compute: a supremum over many $Y$. The next results reduce it to a **limit along one sequence**.

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# Slice the $y$-Axis, Not the $\omega$-Axis




- Cut the $y$-**axis** into slices of height $\frac{1}{2^n}$.
- For each slice, collect all $\omega$ with $X(\omega)$ in that slice. These $\omega$'s may be scattered, and $\Omega$ need not even be numerical.
- On that set, define $X_n =$ the **bottom** of the slice.
- Above height $n$, stop slicing: put a **cap** $X_n = n$ (otherwise infinitely many steps).

**Why slice the $y$-axis?**

$X(\omega)$ is always a real number, so "below" makes sense. Each slice pulls back to a measurable set, because intervals are Borel sets and $X$ is measurable.

# The Construction

## Definition of $X_n$ ($n = 1, 2, 3, \dots$)

For $X \geq 0$ and each $\omega \in \Omega$:

$$X_n(\omega) = \begin{cases} n, & X(\omega) \geq n, \\ \dfrac{k}{2^n}, & \dfrac{k}{2^n} \leq X(\omega) < \dfrac{k+1}{2^n}, \quad k = 0, 1, \dots, n2^n - 1. \end{cases}$$

## Counting

Cases: $k = 0, \dots, n2^n - 1$ plus the cap, so $n2^n + 1$ values $\{0, \frac{1}{2^n}, \frac{2}{2^n}, \dots, n\}$. Finitely many values, so $X_n$ is **simple**.

## Numerical check: $X(\omega) = 2.7$ and $X(\omega) = 0.7$

| $n$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $X_n$ when $X = 0.7$ | 0.5 | 0.5 | 0.625 | 0.6875 |
| $X_n$ when $X = 2.7$ | 1 | 2 | 2.625 | 2.6875 |

Nondecreasing in $n$, and creeping up to $X$.

# Theorem 1 and Proof (Part 1): It Is a Simple Random Variable

## Theorem 1

If $X \geq 0$ is a random variable, then the $X_n$ above are non-negative simple random variables with

$$X_n(\omega) \leq X_{n+1}(\omega) \quad \text{and} \quad X_n(\omega) \leq X(\omega) \quad \text{for all } \omega, n, \qquad X_n(\omega) \longrightarrow X(\omega).$$

We write $X_n \uparrow X$.

## Measurability

Each set defining a value of $X_n$ is an inverse image of a Borel set:

$$\{X_n = \tfrac{k}{2^n}\} = X^{-1}\big([\tfrac{k}{2^n}, \tfrac{k+1}{2^n})\big) \in \mathcal{F}, \qquad \{X_n = n\} = X^{-1}\big([n, \infty)\big) \in \mathcal{F},$$

because $X$ is measurable. So $X_n = \sum_k \frac{k}{2^n} I_{\{X_n = k/2^n\}} + n I_{\{X \geq n\}}$ is a simple random variable.

## Remark

The $X_n$ lie below $X$ by design: we always take the **bottom** of the slice, and the cap $n$ is below $X$ when $X \geq n$.

# Proof (Part 2): Increasing and Convergent

## Increasing: $X_n \le X_{n+1}$

- $X \ge n+1$: $X_{n+1} = n+1 \ge X_n$.
- $n \le X < n+1$: $X_n = n$, and $X_{n+1} = \lfloor 2^{n+1}X \rfloor / 2^{n+1} \ge n$ because $X \ge n$ and $n2^{n+1}$ is an integer.
- $X < n$: $X_n = \lfloor 2^n X \rfloor / 2^n$ and $X_{n+1} = \lfloor 2^{n+1}X \rfloor / 2^{n+1}$. Since $2\lfloor y \rfloor \le \lfloor 2y \rfloor$, we get $X_n \le X_{n+1}$.

## Convergence: $X_n(\omega) \to X(\omega)$

Fix $\omega$ and $\varepsilon > 0$. Take $n > X(\omega)$ and $2^{-n} < \varepsilon$. Then $\omega$ is in a genuine slice, so

$$0 \le X(\omega) - X_n(\omega) < \tfrac{1}{2^n} < \varepsilon.$$

Hence $X_n(\omega) \to X(\omega)$. □

## Why this matters

**Every** non-negative random variable has such a sequence. It is only one example; any other sequence of simple $0 \le X_n \uparrow X$ will do in the next theorem.

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# Theorem 2: $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$

## Theorem 2

Let $X \geq 0$ be a random variable and let $X_n$ be **any** sequence of non-negative simple random variables with $X_n \uparrow X$ pointwise. Then

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X] \qquad (\text{possibly } +\infty).$$

## Consequences

- To compute $\mathbb{E}[X]$: build *any* increasing simple approximation, compute the finite sums $\mathbb{E}[X_n]$, and take a limit.
- The answer does not depend on which approximating sequence you choose.

## Pointwise vs. a.s.

We assume $X_n \leq X_{n+1} \leq X$ and $X_n \to X$ for *all* $\omega$. The proof with "almost all $\omega$" is the same with probability-one bookkeeping.

# Proof, Step 1: The Limit $C$

## The real sequence $\mathbb{E}[X_n]$

Since $X_n \leq X_{n+1}$, monotonicity gives $\mathbb{E}[X_n] \leq \mathbb{E}[X_{n+1}]$. A non-decreasing sequence of real numbers always has a limit in $[0, \infty]$ (finite if bounded, $+\infty$ if not). Call it

$$C = \lim_{n \to \infty} \mathbb{E}[X_n] \in [0, \infty].$$

## What remains

We must show $\mathbb{E}[X] = C$. We prove two inequalities: $\mathbb{E}[X] \geq C$ and $\mathbb{E}[X] \leq C$.

## Step 2: $\mathbb{E}[X] \geq C$

Each $X_n$ is simple with $0 \leq X_n \leq X$, so it is one of the $Y$'s in the supremum defining $\mathbb{E}[X]$. Hence $\mathbb{E}[X] \geq \mathbb{E}[X_n]$ for every $n$, and letting $n \to \infty$,

$$\mathbb{E}[X] \geq C.$$

(If $\mathbb{E}[X]$ were below $C$, the $\mathbb{E}[X_n]$ would eventually jump above it, contradicting that $\mathbb{E}[X]$ is an upper bound.)

# Proof, Step 3: Reduce $\mathbb{E}[X] \le C$ to Simple $Y$

## Reduction

$\mathbb{E}[X]$ is the **least** upper bound of the numbers $\mathbb{E}[Y]$. So $\mathbb{E}[X] \le C$ follows once we show

$$\mathbb{E}[Y] \le C \qquad \text{for every simple } Y \text{ with } 0 \le Y \le X.$$

Indeed, then $C$ is an upper bound for all the $\mathbb{E}[Y]$, and the least upper bound cannot exceed it.

## Setup

If $C = \infty$ there is nothing to prove, so assume $C < \infty$. Take simple $Y$ with $0 \le Y \le X$, in canonical form

$$Y = \sum_{i=1}^{m} y_i \, I_{B_i}, \qquad B_i = \{Y = y_i\}.$$

Fix $\varepsilon \in (0, 1]$.

## The trick

We cannot compare $X_n$ and $Y$ directly: $X_n$ may sit slightly below $Y$. So we **shrink** $Y$ **a little**, to $(1-\varepsilon)Y$, and cut it off where $X_n$ is too small. The shrunk copy fits under $X_n$.

# Step 3: The Shrunk Copy $Y_{n,\varepsilon}$

## Definition

$$Y_{n,\varepsilon} = (1-\varepsilon)\, Y\, I_{\{X_n \geq (1-\varepsilon)Y\}}.$$

## It lies below $X_n$

- Where $X_n < (1-\varepsilon)Y$: $Y_{n,\varepsilon} = 0 \leq X_n$.
- Where $X_n \geq (1-\varepsilon)Y$: $Y_{n,\varepsilon} = (1-\varepsilon)Y \leq X_n$.

So $0 \leq Y_{n,\varepsilon} \leq X_n$, and by monotonicity $\mathbb{E}[Y_{n,\varepsilon}] \leq \mathbb{E}[X_n] \leq C$.

## Its expectation

A product of indicators is an indicator of the intersection. Put

$$B_{n,i,\varepsilon} = \{X_n \geq (1-\varepsilon)Y\} \cap B_i \in \mathcal{F}.$$

Then $Y_{n,\varepsilon} = (1-\varepsilon)\sum_i y_i I_{B_{n,i,\varepsilon}}$ is simple, and

$$\mathbb{E}[Y_{n,\varepsilon}] = (1-\varepsilon)\sum_{i=1}^{m} y_i\, \mathbb{P}(B_{n,i,\varepsilon}).$$

# Step 3: Letting $n \to \infty$, then $\varepsilon \to 0$

## Key claim (proved on the next two slides)

For each fixed $i$ and $\varepsilon$: $B_{n,i,\varepsilon} \uparrow B_i$ as $n \to \infty$.

## Use continuity of probability from below

$\mathbb{P}(B_{n,i,\varepsilon}) \uparrow \mathbb{P}(B_i)$. Since the sum over $i$ is finite,

$$\mathbb{E}[Y_{n,\varepsilon}] \longrightarrow (1-\varepsilon)\sum_{i=1}^{m} y_i \mathbb{P}(B_i) = (1-\varepsilon)\,\mathbb{E}[Y].$$

## Combine

$\mathbb{E}[Y_{n,\varepsilon}] \le C$ for every $n$, so in the limit

$$(1-\varepsilon)\,\mathbb{E}[Y] \le C.$$

This holds for every $\varepsilon \in (0,1]$. Letting $\varepsilon \downarrow 0$ gives $\mathbb{E}[Y] \le C$ (here $\mathbb{E}[Y] < \infty$ since $Y$ is simple). □

## Conclusion of Theorem 2

$\mathbb{E}[X] \ge C$ (Step 2) and $\mathbb{E}[X] \le C$ (Step 3), so $\mathbb{E}[X] = C = \lim_n \mathbb{E}[X_n]$.

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# Claim: $B_{n,i,\varepsilon} \uparrow B_i$ (Part 1: Increasing)

## Recall

$B_{n,i,\varepsilon} = \{\omega : \ X_n(\omega) \geq (1-\varepsilon)Y(\omega)\} \cap B_i$, and we need (a) $B_{n,i,\varepsilon} \subseteq B_{n+1,i,\varepsilon}$ and (b) $\bigcup_n B_{n,i,\varepsilon} = B_i$.

## (a) Increasing in $n$

Since $X_n \leq X_{n+1}$, it is **easier** for $X_{n+1}$ to satisfy the inequality:

$$\{X_n \geq (1-\varepsilon)Y\} \subseteq \{X_{n+1} \geq (1-\varepsilon)Y\}.$$

Intersect both sides with $B_i$.

## (b, easy half) $\bigcup_n B_{n,i,\varepsilon} \subseteq B_i$

Each $B_{n,i,\varepsilon}$ is by definition a subset of $B_i$, so the union is too.

# Claim (Part 2): Every $\omega \in B_i$ Eventually Enters

**Take $\omega \in B_i$**

We must find some $N$ with $X_n(\omega) \geq (1-\varepsilon)Y(\omega)$ for all $n \geq N$. Then $\omega \in B_{n,i,\varepsilon}$ for $n \geq N$.

**Case 1: $Y(\omega) = 0$**

Then the inequality reads $X_n(\omega) \geq 0$, which is always true. Take $N = 1$.

**Case 2: $Y(\omega) > 0$**

We know $Y(\omega) \leq X(\omega)$, so $(1-\varepsilon)Y(\omega) < Y(\omega) \leq X(\omega)$, with a strict first inequality because $\varepsilon > 0$ and $Y(\omega) > 0$. Since $X_n(\omega) \to X(\omega)$, the sequence is eventually above any number strictly below its limit. So for $n \geq N$,

$$X_n(\omega) > (1-\varepsilon)Y(\omega).$$

**Why we shrank $Y$**

If $\varepsilon = 0$ we could have $X_n(\omega) < Y(\omega) \leq X(\omega)$ for *all* $n$ (approaching from below). The room $(1-\varepsilon)$ creates a **strict** gap, which a convergent sequence must eventually cross.

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# Example: $\mathbb{E}[-2\ln\omega]$ on $(0,1]$

## Setup

$\Omega = (0,1]$, Lebesgue measure as $\mathbb{P}$, and $X(\omega) = -2\ln\omega \geq 0$. Use our construction $X_n$ with $h = 2^{-n}$.

## Pull back the slices

$X(\omega) \geq a \iff \omega \leq e^{-a/2}$. So with $r = e^{-h/2}$,

$$\{\tfrac{k}{2^n} \leq X < \tfrac{k+1}{2^n}\} = \big(r^{k+1},\, r^k\big], \quad \text{length } r^k(1-r), \qquad \{X \geq n\} = (0,\, e^{-n/2}].$$

## Check $n = 1$ ($h = \frac{1}{2}$, values $0, \frac{1}{2}, 1$)

$$\mathbb{E}[X_1] = 0 + \tfrac{1}{2}\big(e^{-1/4} - e^{-1/2}\big) + 1 \cdot e^{-1/2} = \tfrac{1}{2}\big(e^{-1/4} + e^{-1/2}\big).$$

# Example: General $n$ and the Limit

## Compute $\mathbb{E}[X_n]$

With $N = n2^n$, $r = e^{-h/2}$ and $r^N = e^{-n/2}$:

$$\mathbb{E}[X_n] = \sum_{k=0}^{N-1} kh\, r^k(1-r) + n\, r^N.$$

Regrouping the telescoping terms (each $k\, r^k(1-r)$ gives $kr^k - kr^{k+1}$) leaves

$$\mathbb{E}[X_n] = h\sum_{k=1}^{N} r^k = \frac{h\, r\,(1-r^N)}{1-r} = \frac{2^{-n}\, e^{-2^{-n-1}}\,(1-e^{-n/2})}{1-e^{-2^{-n-1}}}.$$

## Limit as $n \to \infty$

$r \to 1$ and $1 - e^{-n/2} \to 1$. For the rest, with $t = 2^{-n-1} \to 0$: $\dfrac{2^{-n}}{1-e^{-t}} = \dfrac{2t}{1-e^{-t}} \to 2$ (since $1 - e^{-t} \sim t$; l'Hôpital gives the same). Hence

$$\lim_{n\to\infty} \mathbb{E}[X_n] = 2 \quad \Longrightarrow \quad \mathbb{E}[-2\ln\omega] = 2.$$

## Sanity check

$\int_0^1 -2\ln\omega\, d\omega = 2$. Same answer, but this route needed no integral, only sums and a limit.

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# Summary

## Key results

| Concept | Content |
|---|---|
| Definition ($X \geq 0$) | $\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : \ Y \text{ simple}, \ 0 \leq Y \leq X\} \in [0, \infty]$ |
| Approximation | $X_n = \frac{k}{2^n}$ on $\{\frac{k}{2^n} \leq X < \frac{k+1}{2^n}\}$, $X_n = n$ on $\{X \geq n\}$; $X_n \uparrow X$ |
| Measurability | slices are $X^{-1}(\text{Borel interval}) \in \mathcal{F}$ |
| Theorem 2 | $0 \leq X_n \uparrow X$ simple $\Rightarrow \mathbb{E}[X_n] \uparrow \mathbb{E}[X]$ |
| Proof idea | $C = \lim \mathbb{E}[X_n]$; $\mathbb{E}[X] \geq C$ trivially; for $\mathbb{E}[X] \leq C$ use $Y_{n,\varepsilon} = (1-\varepsilon) Y I_{\{X_n \geq (1-\varepsilon)Y\}}$ and continuity from below |
| Example | $\mathbb{E}[-2 \ln \omega] = 2$ via $\mathbb{E}[X_n] = h\, r(1 - r^N)/(1 - r)$ |

# Exercises

1. For $X(\omega) = \omega$ on $[0, 1]$ with Lebesgue measure, write $X_1, X_2$ explicitly and compute $\mathbb{E}[X_1], \mathbb{E}[X_2]$. Guess $\lim \mathbb{E}[X_n]$.
2. For $X = 3.3$ (constant), list $X_1, \ldots, X_5$. Do they increase to 3.3?
3. Show that $X_n \le X_{n+1}$ in the case $X < n$ using $2\lfloor y \rfloor \le \lfloor 2y \rfloor$.
4. Why is a cap at height $n$ needed in the construction? What goes wrong for unbounded $X$ without it?
5. Show that for simple $X$ the supremum definition returns the old $\mathbb{E}[X]$.
6. In Step 3, why is $B_{n,i,\varepsilon} \in \mathcal{F}$? (Hint: $X_n$ and $Y$ are simple.)
7. What would fail in "Case 2" if we used $\varepsilon = 0$? Give an example sequence.
8. Check the formula $\mathbb{E}[X_n] = h \sum_{k=1}^{N} r^k$ for $n = 1$ against the direct computation.
9. Let $X$ take the value $+\infty$ with probability 0 only. Explain why we do not lose generality by working with real-valued $X$.

# Outlook

## Next lesson

Expectation for **general** random variables: split $X = X^+ - X^-$ with $X^\pm \geq 0$ and define $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$ when this makes sense. We then introduce the **integral** notation $\int X \, d\mathbb{P}$, and integration against general measures $d\mu$.

## Looking ahead

Later we show that this definition agrees with the density-based formulas you know from probability and statistics.